

HEISENBERG GROUP – BASED DIGITAL SIGNATURE FOR SECURE COMMUNICATION

A Project Report (20A05801) submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY

ANANTAPUR, ANANTAPURAMU

In partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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2025-2026



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Certificate

This is to certify that the Project Report (20A05801) entitled
**HEISENBERG GROUP – BASED DIGITAL
SIGNATURE FOR SECURE COMMUNICATION**

is the bonafide work done and
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ACKNOWLEDGEMENT

Any achievement, be it scholastic or otherwise does not depend solely on the individual effort but on the guidance, encouragement and cooperation of intellectuals, elders and friends. We would like to take this opportunity to thank them all.

We feel ourselves honoured for placing our warm salutation to **THE MANAGEMENT** Mother Theresa Institute of Engineering and Technology, Palamaner, which gave us the opportunity to obtain a strong base in B. Tech and profound knowledge.

We express our sincere thanks to **Dr. M.LAKSHMIKANTHA REDDY**, M.Tech, Ph.D, our beloved Principal for his encouragement and suggestions during our course of study.

With deep sense of gratitude we acknowledge **Dr. P.C. PRABHU KUMAR**, M.Tech, Ph.D Head of the Dept., Computer Science & Engineering, for his valuable support and help in processing our Socially Relevant Project.

We also express sincere thanks to our Project Coordinator **MRS. P. POORNIMA**, M.Tech, (Ph.D) Assistant Professor in Department of Computer Science and Engineering, for encouraging us in doing this Project.

We whole-heartedly express gratitude and regards to our guide **Mrs.. V. SUJANA**, M.Tech.

Assistant Professor in Department of Computer Science and Engineering, for constant support and encouragement.

Finally, we would like to express our sincere thanks to **Faculty Members** of C.S.E Department, and Lab Technicians, Friends & Family members, who all have motivated and helped us to do this Socially Relevant Project.

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ABSTRACT

In this project, we present a secure cloud-based data storage solution that employs **Edwards-Curve Digital Signature Algorithm (EdDSA)** to ensure the integrity and authenticity of stored data. Users can upload various types of data, including files, videos, and music, to the cloud. To secure communication between the user and the server, the system utilizes **EdDSA**, a modern and efficient cryptographic algorithm, which provides enhanced security, data integrity, and non-repudiation. This approach leverages elliptic curve cryptography to protect against unauthorized access and safeguard data against emerging security threats, including those posed by quantum computing. **EdDSA** offers a faster, more secure, and scalable solution compared to traditional digital signature schemes, making it highly suitable for cloud data storage systems. The proposed system provides a quantum-resilient solution for cloud storage, ensuring that sensitive data remains protected against both current and future cryptographic challenges.

KEYWORDS: Cloud Storage, EdDSA, Digital Signatures, Data Integrity, Authentication, Cryptography, Non-repudiation, Secure Communication, Quantum Security, Data Protection.

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1. INTRODUCTION

1.1 Motivation:

With the increasing reliance on cloud storage for sensitive data, ensuring robust security is critical. Traditional cryptographic methods like RSA and ECDSA are becoming vulnerable to emerging threats, particularly quantum computing. **EdDSA**, based on elliptic curve cryptography, offers enhanced security, smaller key sizes, and faster performance compared to conventional methods. It is also more resistant to quantum attacks, making it a future-proof solution for cloud data protection. This project aims to provide a secure, scalable, and efficient cryptographic solution using **EdDSA** to address current and future security challenges in cloud storage.

1.2 Problem Statement:

As cloud storage becomes the primary solution for managing sensitive data, ensuring its security and integrity has become a major concern. Traditional cryptographic methods like RSA and ECDSA are increasingly vulnerable to quantum computing threats, which can compromise data confidentiality and integrity. The challenge lies in developing a secure, scalable, and future-proof cloud storage system that can withstand evolving cryptographic threats, including quantum computing. This project addresses these concerns by implementing **EdDSA**, a quantum-resistant cryptographic algorithm, to ensure secure data storage, integrity, and communication in the cloud.

1.3 Objective of the Project:

The objective of this project is to develop a secure cloud storage system that integrates **EdDSA (Edwards-Curve Digital Signature Algorithm)** to ensure data integrity, authenticity, and confidentiality. The system aims to provide secure communication and data storage by utilizing **EdDSA** digital signatures, ensuring resistance against quantum computing threats, and making it future-proof. Additionally, the project focuses on guaranteeing non-repudiation and integrity of the uploaded data, developing a scalable solution that can efficiently handle increasing data

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volumes and user load, and providing a user-friendly interface for easy data upload, retrieval, and management within the cloud.

1.4 Scope:

This project focuses on developing a secure cloud storage system that integrates **EdDSA (Edwards-Curve Digital Signature Algorithm)** to enhance data security. It includes features for users to upload, store, and retrieve various types of data, such as documents, videos, and music, to the cloud. The system will ensure the integrity and confidentiality of the data through **EdDSA** digital signatures, protecting it from unauthorized access and tampering. The project also addresses the growing concern of quantum computing threats, ensuring the system remains secure in a post-quantum environment. Additionally, it emphasizes scalability, allowing the system to handle large volumes of data and users while maintaining high performance. The user interface will be designed for ease of use, enabling seamless interaction with the cloud storage system.

1.5 Project Introduction:

In today's digital world, cloud storage has become a fundamental service for individuals and organizations alike, allowing easy access to data and the ability to share it across multiple devices. It plays a crucial role in storing sensitive and critical data, from personal documents to business information. However, as the volume of sensitive data stored in the cloud grows, the need for robust security mechanisms becomes even more critical.

Traditional encryption techniques, such as RSA and ECDSA, have been widely adopted to protect data in the cloud. While these methods provide a certain level of security, they are increasingly vulnerable to the rise of quantum computing. Quantum computers, with their ability to solve complex mathematical problems much faster than classical computers, can potentially break the security of widely used cryptographic systems, posing a significant threat to data confidentiality and integrity.

This project aims to tackle these challenges by developing a **quantum-resistant** cloud storage system using **EdDSA (Edwards-Curve Digital Signature Algorithm)**. **EdDSA** is a modern elliptic curve-based digital signature scheme that offers several advantages over traditional

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cryptographic algorithms. It provides better security, smaller key sizes, and faster performance, making it more efficient for securing cloud-based data. Furthermore, **EddSA** is resistant to certain cryptographic attacks, including those posed by quantum computing, making it a suitable choice for future-proofing cloud storage systems.

The proposed system leverages **EddSA** to provide secure communication and data storage by signing and verifying files and messages. This ensures that the data remains untampered with and authentic. The system will enable users to upload, store, and retrieve various types of files, including documents, images, videos, and music, while ensuring their confidentiality and integrity through **EddSA** digital signatures. Additionally, the system will offer features such as **non-repudiation**, meaning that once a file is uploaded and signed, it cannot be denied or altered by the sender.

The scalability of the system is also a key consideration. As the demand for cloud storage grows, the system is designed to handle increasing data volumes and user traffic efficiently while maintaining high levels of security and performance. The user-friendly interface will allow users to seamlessly interact with the system, ensuring that they can easily upload, view, and manage their files in the cloud.

By incorporating **EddSA**, this project provides a secure, scalable, and quantum-resistant solution for cloud data storage, ensuring that sensitive data remains protected in the face of evolving cryptographic threats. The system not only addresses current security needs but also prepares for future challenges posed by emerging technologies, ensuring long-term security and reliability for cloud-based data storage.

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2. LITERATURE SURVEY

2.1 Related Work:

[1] Dongsheng Li, Qingxu Li, and Shanshan Li, “A Survey on Post-Quantum Cryptography: Algorithms and Implementations.” IEEE Access, 2021.

This paper provides an in-depth survey of post-quantum cryptography, focusing on algorithms that are resistant to quantum computing attacks. The authors explore various cryptographic methods, such as lattice-based cryptography, code-based cryptography, and multivariate quadratic equations, which are believed to be secure against quantum algorithms like Shor’s algorithm. The paper discusses the strengths and limitations of these methods, as well as their potential for practical implementation in the future.

[2] M. W. J. F. van der Linden, “Quantum-Resistant Cryptography: A Survey on the Security of Digital Signature Schemes.” Journal of Cryptographic Engineering, 2020.

This survey focuses on the security of digital signature schemes in a post-quantum environment. The paper examines quantum-resistant alternatives to traditional schemes like RSA and ECDSA, including hash-based signatures and lattice-based cryptographic schemes. It analyzes the trade-offs between security and performance in the context of quantum computing, providing valuable insights into the design and implementation of secure digital signature systems that can withstand quantum threats.

[3] David J. Bernstein, Tanja Lange, and Chris Peikert, “Post-Quantum Cryptography: A New Generation of Secure Systems.” Springer, 2018.

This book offers a comprehensive overview of post-quantum cryptography, presenting a variety of cryptographic primitives and schemes designed to resist quantum attacks. It explores the latest advancements in quantum-safe algorithms, including lattice-based cryptography, code-based cryptography, and multivariate polynomial equations, and discusses their potential applications in areas such as digital signatures, key exchange, and encryption.

[4] E. B. Tamassia, P. Q. Nguyen, and S. M. Yiu, “Heisenberg Group and its Applications to Cryptography.” Journal of Mathematical Cryptology, 2019.

This paper focuses on the Heisenberg group and its application to cryptography, particularly for secure communication and data integrity. The authors explore the non-commutative properties of the Heisenberg group and how these properties can be leveraged to develop robust cryptographic systems that are resistant to classical and quantum attacks. The paper also discusses potential use cases in secure

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data transmission and digital signature generation, providing a foundation for quantum-resistant cryptographic methods.

[5] **D. P. Cohn, T. W. L. Chan, and P. Y. Cheng, “Quantum-Safe Cryptography: A Survey on Advanced Signature Schemes for Post-Quantum Security.” IEEE Transactions on Information Forensics and Security, 2022.**

This paper surveys advanced signature schemes for post-quantum security, focusing on the need for quantum-safe alternatives to traditional signature algorithms. It reviews several cryptographic models, including those based on hash functions, lattices, and other mathematical constructs. The paper highlights the potential of these advanced schemes for providing secure, scalable, and efficient solutions to the digital signature problem in cloud storage and other cryptographic applications, making it highly relevant to the context of the proposed project.

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3. SYSTEM ANALYSIS

3.1 Existing System

Traditional cloud storage systems use cryptographic methods like **RSA**, **ECDSA**, and **AES** to secure data. RSA and ECDSA, though widely used, are vulnerable to quantum computing threats, as quantum algorithms can efficiently break these encryption methods. AES, while secure, may require larger key sizes to withstand quantum attacks, affecting performance. These traditional systems struggle with scalability and efficiency, especially with large data volumes, as they rely on complex encryption and signature verification processes. These limitations highlight the need for a more secure and future-proof solution, such as **EdDSA (Edwards-Curve Digital Signature Algorithm)**, which offers better security, faster performance, and quantum resistance.

3.2 Disadvantages

- **Not Fully Quantum-Resistant:** **EdDSA** provides better security than traditional algorithms but is not entirely immune to quantum computing threats.
- **Implementation Complexity:** **EdDSA** may be more complex to implement compared to simpler cryptographic methods, requiring more development time.
- **Computational Overhead:** While efficient, **EdDSA** still adds computational overhead compared to basic encryption methods.
- **Legacy System Compatibility:** Integrating **EdDSA** into older systems may present compatibility issues.
- **Limited Adoption:** **EdDSA** is relatively new, so support in some platforms and libraries may be limited.

3.3 Proposed System

The proposed system aims to secure cloud storage by integrating **EdDSA (Edwards-Curve Digital Signature Algorithm)** for digital signatures, ensuring data integrity, authenticity, and confidentiality. Users will be able to upload, store, and retrieve various types of data securely, with **EdDSA** providing quantum-resistant protection against tampering. The system will sign and verify data using **EdDSA**, ensuring non-repudiation and preventing unauthorized access.

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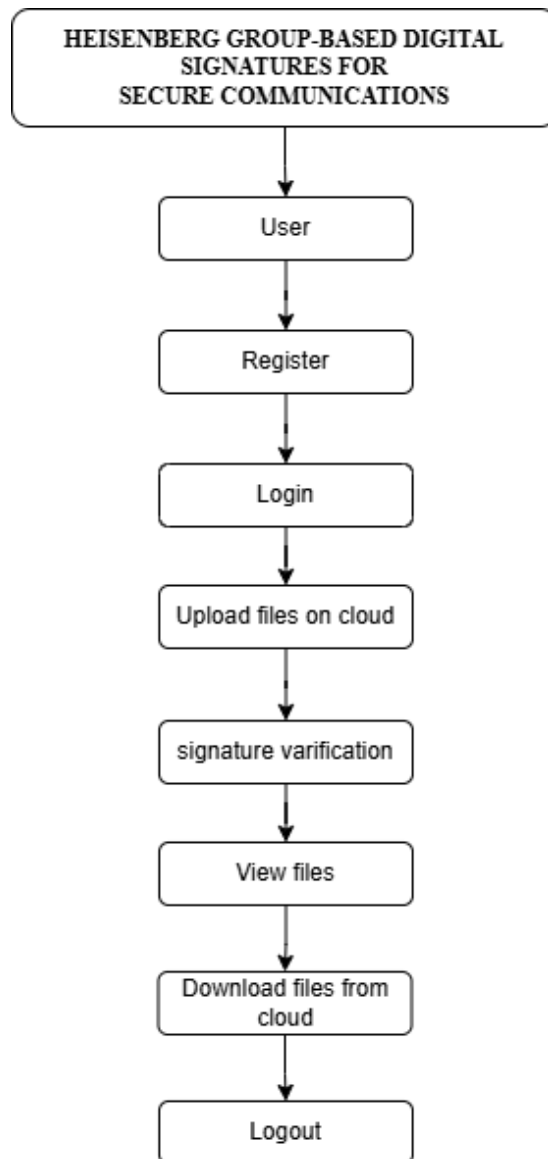
It will also offer scalable, high-performance security, with a user-friendly interface for seamless file management. This approach provides a future-proof, secure solution for cloud storage, addressing both current and emerging cryptographic threats.

3.4 Advantages

- **Enhanced Security:** By leveraging **EdDSA (Edwards-Curve Digital Signature Algorithm)**, the system offers robust cryptographic security, ensuring data integrity, authenticity, and confidentiality.
- **Quantum Resistance:** **EdDSA** is more resistant to quantum computing threats compared to traditional algorithms like RSA and ECDSA, making the system future-proof.
- **Faster Performance:** **EdDSA** provides faster signature generation and verification than older algorithms, enhancing system efficiency.
- **Smaller Key Sizes:** The use of elliptic curve cryptography ensures smaller key sizes for the same level of security, improving computational efficiency.
- **Non-Repudiation:** The system ensures that once data is uploaded and signed, it cannot be tampered with or denied by the sender, ensuring trust in the data.
- **Scalability:** The system is designed to efficiently handle large volumes of data and users, making it suitable for growing cloud storage needs.

3.5 Work Flow of Proposed system

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4. REQUIREMENT ANALYSIS

4.1 Functional and non-functional requirements

Requirement's analysis is very critical process that enables the success of a system or software project to be assessed. Requirements are generally split into two types: Functional and non-functional requirements.

Functional Requirements: These are the requirements that the end user specifically demands as basic facilities that the system should offer. All these functionalities need to be necessarily incorporated into the system as a part of the contract. These are represented or stated in the form of input to be given to the system, the operation performed and the output expected. They are basically the requirements stated by the user which one can see directly in the final product, unlike the non-functional requirements.

Examples of functional requirements:

- 1) Authentication of user whenever he/she logs into the system
- 2) System shutdown in case of a cyber-attack
- 3) A verification email is sent to user whenever he/she register for the first time on some software system.

Non-functional requirements: These are basically the quality constraints that the system must satisfy according to the project contract. The priority or extent to which these factors are implemented varies from one project to other. They are also called non-behavioral requirements.

They basically deal with issues like:

- Portability
- Security
- Maintainability
- Reliability
- Scalability
- Performance
- Reusability
- Flexibility

Examples of non-functional requirements:

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- 1) Emails should be sent with a latency of no greater than 12 hours from such an activity.
- 2) The processing of each request should be done within 10 seconds
- 3) The site should load in 3 seconds whenever of simultaneous users are > 10000

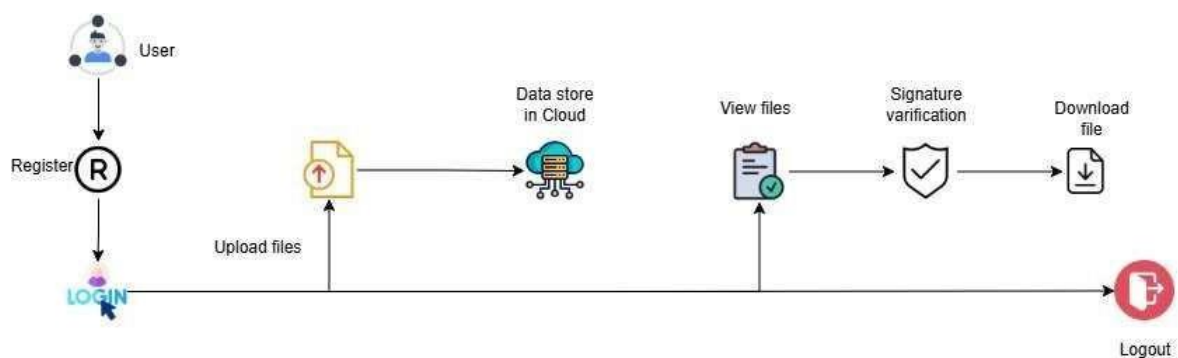
4.2 Hardware Requirements

Processor	- I3/Intel Processor
Hard Disk	- 160GB
Key Board	- Standard Windows Keyboard
Mouse	- Two or Three Button Mouse
Monitor	- SVGA
RAM	- 8GB

4.3 Software Requirements:

Operating System	: Windows 7/8/10
Server side Script	: HTML, CSS, Bootstrap & JS
Programming Language	: Python
Libraries	: Django, Numpy
IDE/Workbench	: VS Code
Technology	: Python 3.6+
Server Deployment	: Xampp Server
Database	: MySQL

4.4 Architecture:



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5. SYSTEM DESIGN

5.1 Introduction of Input Design:

In an information system, input is the raw data that is processed to produce output. During the input design, the developers must consider the input devices such as PC, MICR, OMR, etc.

Therefore, the quality of system input determines the quality of system output. Well-designed input forms and screens have following properties –

- It should serve specific purpose effectively such as storing, recording, and retrieving the information.
- It ensures proper completion with accuracy.
- It should be easy to fill and straightforward.
- It should focus on user's attention, consistency, and simplicity.
- All these objectives are obtained using the knowledge of basic design principles regarding –
 - What are the inputs needed for the system?
 - How end users respond to different elements of forms and screens.

Objectives for Input Design:

The objectives of input design are –

- To design data entry and input procedures
- To reduce input volume
- To design source documents for data capture or devise other data capture methods
- To design input data records, data entry screens, user interface screens, etc.
- To use validation checks and develop effective input controls.

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5.2 Output Design:

The design of output is the most important task of any system. During output design, developers identify the type of outputs needed, and consider the necessary output controls and prototype report layouts.

The objectives of input design are:

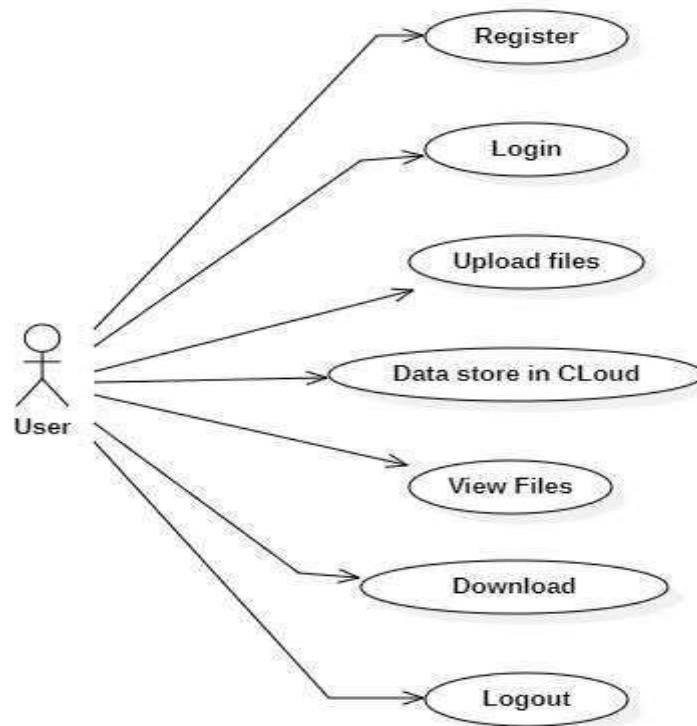
- To develop output design that serves the intended purpose and eliminates the production of unwanted output.
- To develop the output design that meets the end user's requirements.
- To deliver the appropriate quantity of output.
- To form the output in appropriate format and direct it to the right person.
- To make the output available on time for making good decisions.

5.3 UML Diagrams:

5.3.1 Use Case Diagram:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

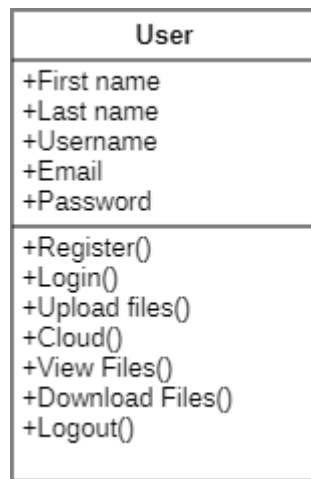
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5.3.2 Class Diagram:

In software engineering, a class diagram in the Unified Modelling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

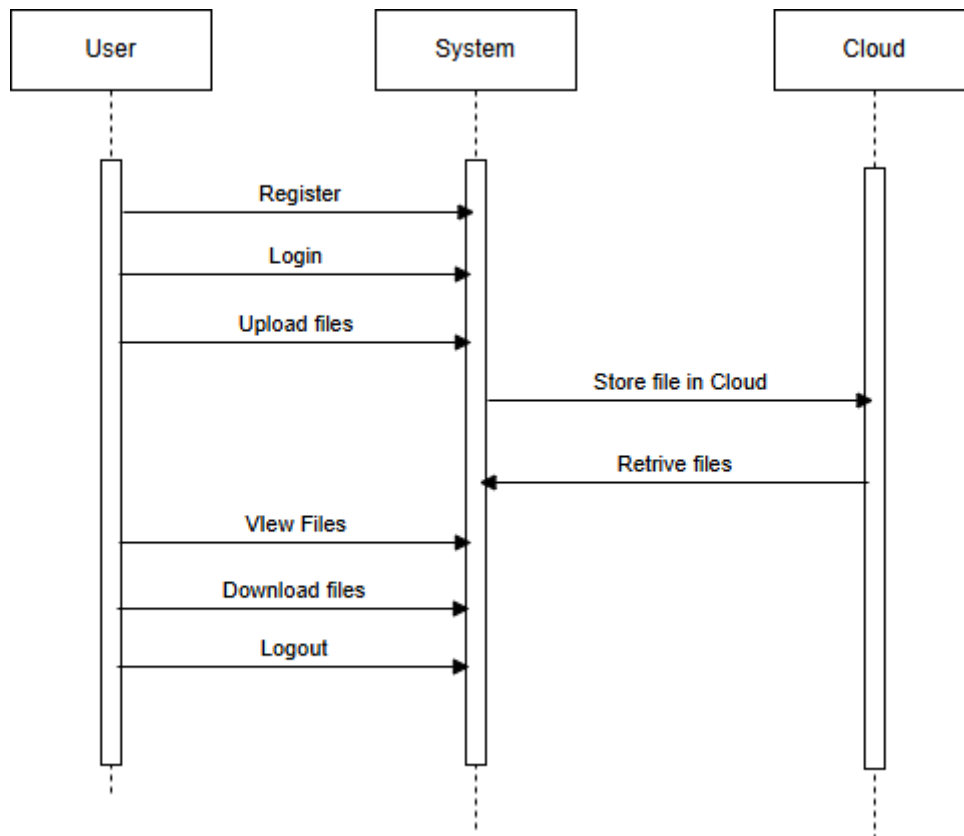
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5.3.3 Sequence Diagram:

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

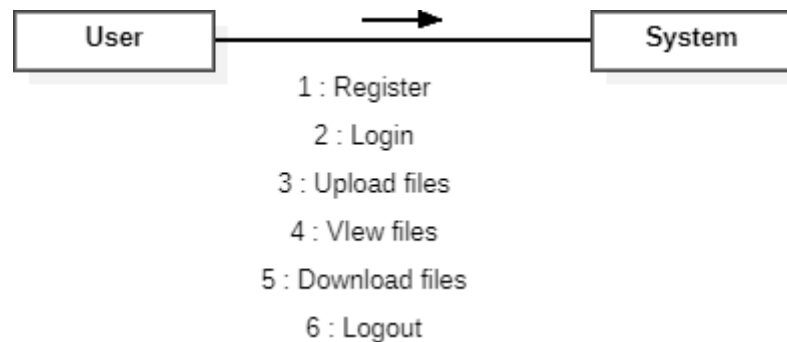
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5.3.4 Collaboration Diagram:

In collaboration diagram the method call sequence is indicated by some numbering technique as shown below. The number indicates how the methods are called one after another. We have taken the same order management system to describe the collaboration diagram. The method calls are similar to that of a sequence diagram. But the difference is that the sequence diagram does not describe the object organization whereas the collaboration diagram shows the object organization.

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5.3.5 Deployment Diagram

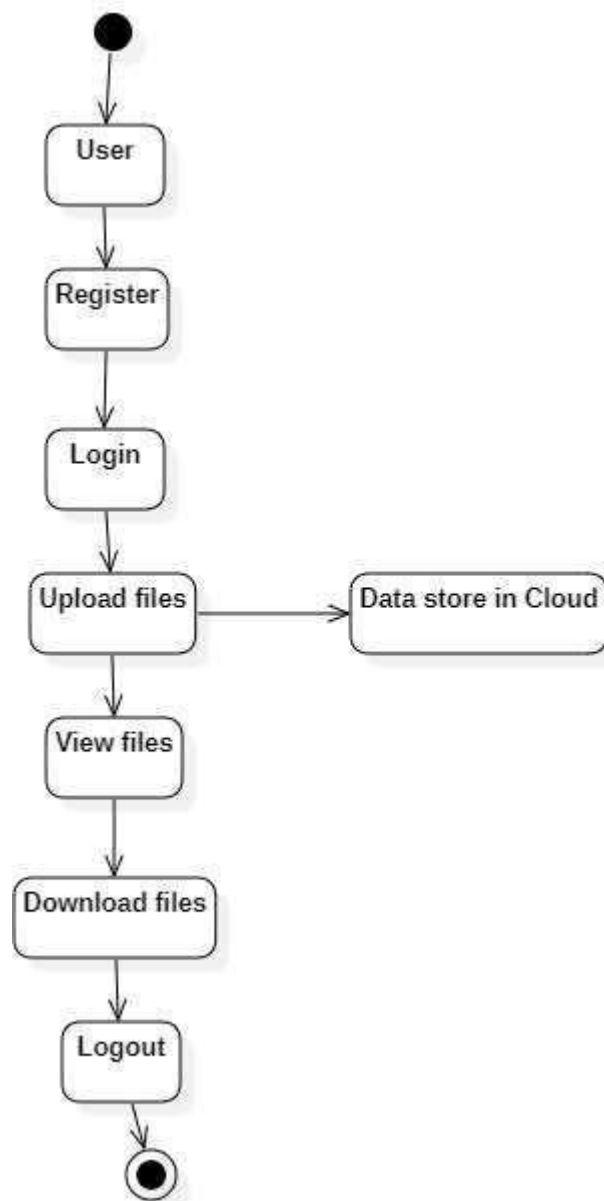
Deployment diagram represents the deployment view of a system. It is related to the component diagram. Because the components are deployed using the deployment diagrams. A deployment diagram consists of nodes. Nodes are nothing but physical hardware's used to deploy the application.



5.3.6 Activity Diagram:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modelling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

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5.3.7 Component Diagram:

A component diagram, also known as a UML component diagram, describes the organization and wiring of the physical components in a system. Component diagrams are often drawn to help model implementation details and double-check that every aspect of the system's required functions is covered by planned development.

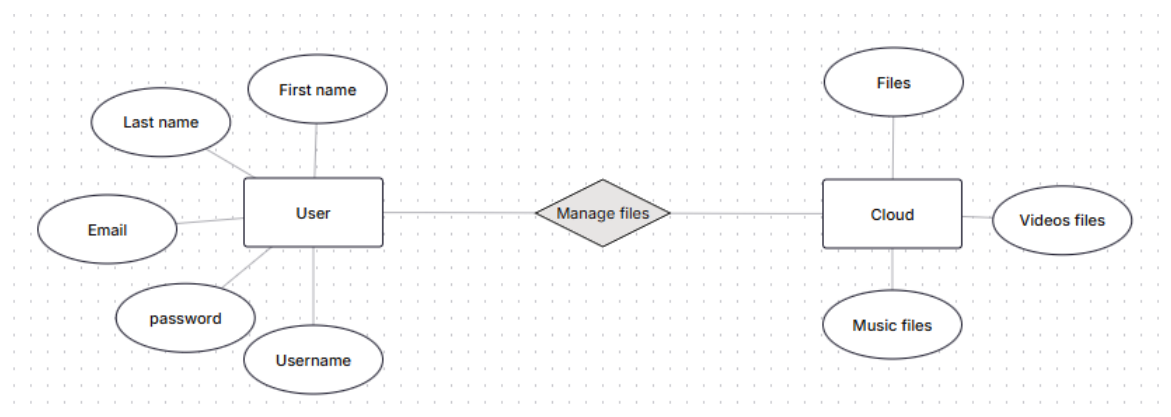
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5.3.8 ER Diagram:

An Entity–relationship model (ER model) describes the structure of a database with the help of a diagram, which is known as Entity Relationship Diagram (ER Diagram). An ER model is a design or blueprint of a database that can later be implemented as a database. The main components of E-R model are: entity set and relationship set.

An ER diagram shows the relationship among entity sets. An entity set is a group of similar entities and these entities can have attributes. In terms of DBMS, an entity is a table or attribute of a table in database, so by showing relationship among tables and their attributes, ER diagram shows the complete logical structure of a database. Let's have a look at a simple ER diagram to understand this concept.

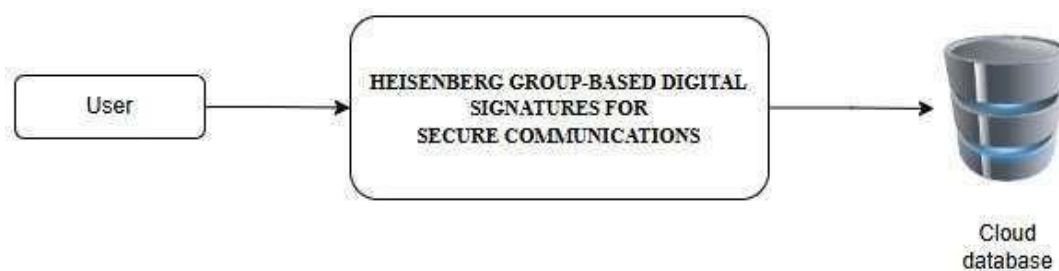


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5.4 DFD Diagram:

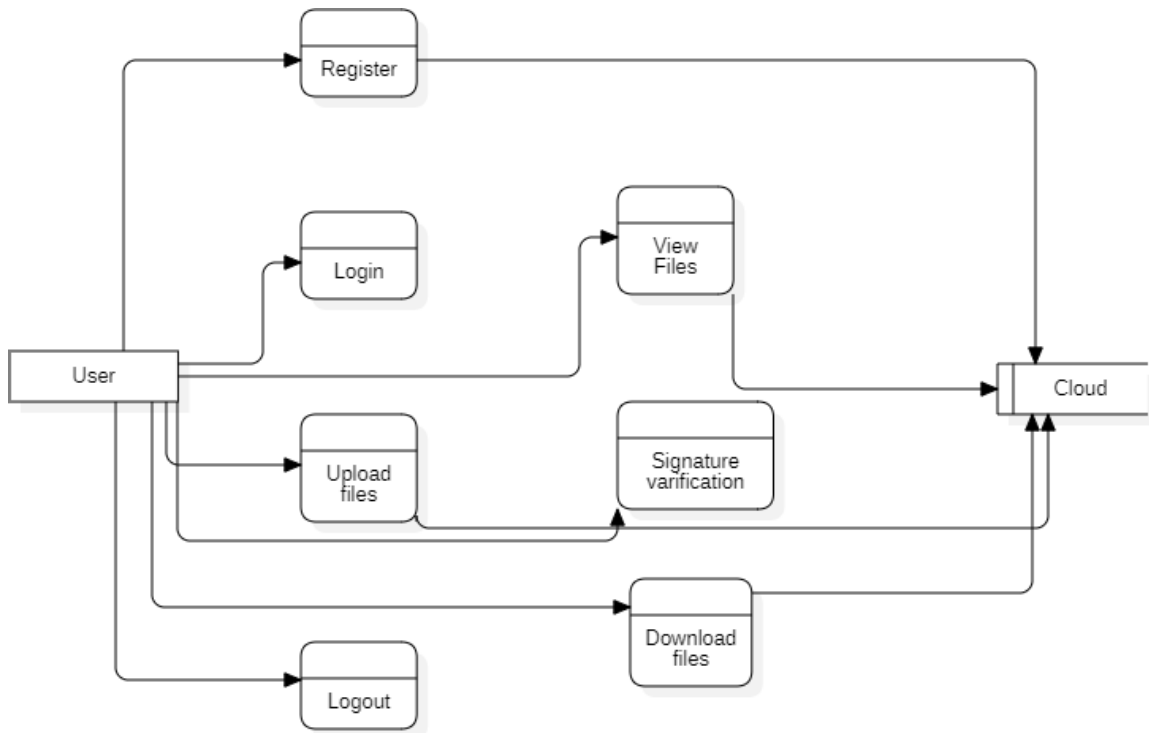
A Data Flow Diagram (DFD) is a traditional way to visualize the information flows within a system. A neat and clear DFD can depict a good amount of the system requirements graphically. It can be manual, automated, or a combination of both. It shows how information enters and leaves the system, what changes the information and where information is stored. The purpose of a DFD is to show the scope and boundaries of a system as a whole. It may be used as a communications tool between a systems analyst and any person who plays a part in the system that acts as the starting point for redesigning a system.

Level 0:



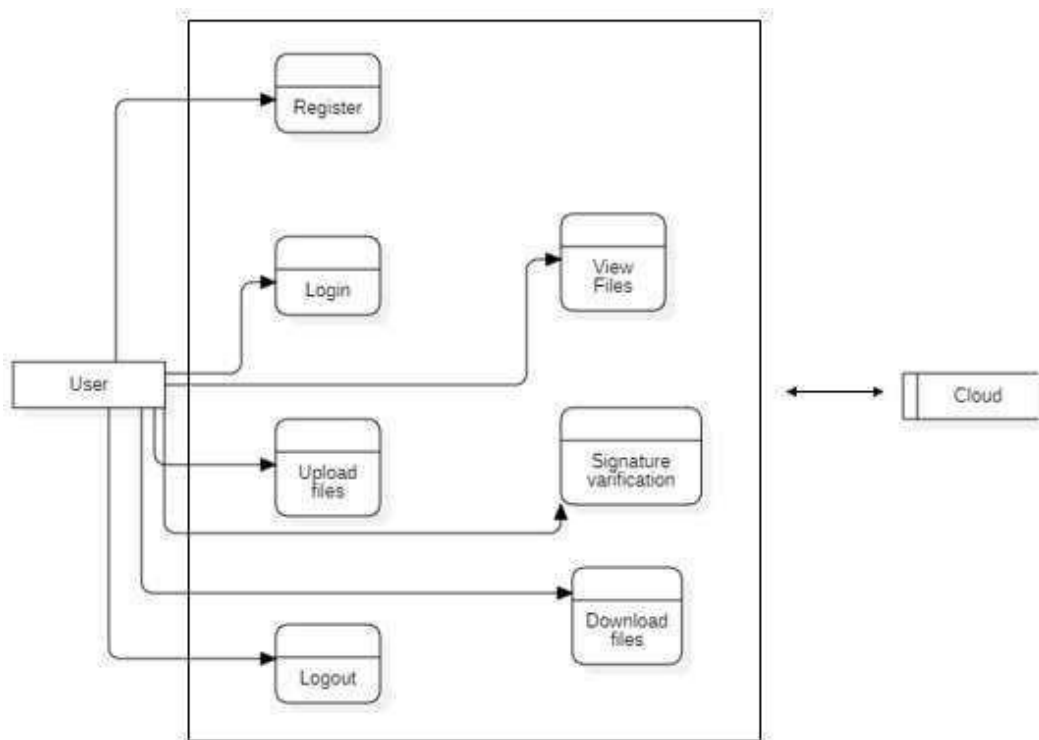
Level 1 Diagram:

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Level 2 Diagram:

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6. MODULES/IMPLEMENTATION

6.1 MODULES:

1. User Module

Functionality: The initial entry point of the system where the user accesses the application.

2. Register Module

Functionality: Allows the user to create a new account by providing necessary details (e.g., username, email, password). This module stores user data in the system's database.

3. Login Module

Functionality: Authenticates users using their registered credentials (username/email and password). After successful authentication, the user is granted access to the system.

4. Upload Files on Cloud Module

Functionality: Enables users to upload various types of files (documents, images, videos, music, etc.) to the cloud storage. The files are securely stored after being signed using Heisenberg group-based digital signatures for security.

5. Signature Verification Module

Functionality: After uploading files, this module verifies the Heisenberg group-based digital signature to ensure the integrity and authenticity of the uploaded files. It checks whether the file has been tampered with during the upload process.

6. View Files Module

Functionality: Allows users to view the list of files that have been uploaded to the cloud. This module provides a user interface to view file metadata or contents (depending on the file type).

7. Download Files from Cloud Module

Functionality: Allows users to download files from the cloud after verifying their authenticity and integrity through digital signatures. This module ensures that the downloaded files are the same as the uploaded ones.

8. Logout Module

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Functionality: Ends the user session and securely logs the user out of the system, ensuring that no unauthorized access occurs after the session ends.

6.2 Results:

HOME:



Register:



Login:

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User Dashboard:



Upload files:

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MY files:



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7. Methodology

7.1 EdDSA (Edwards-Curve Digital Signature Algorithm)

An **algorithm** is a finite, step-by-step procedure or set of rules designed to perform a specific task or solve a problem. It takes an input, processes it through a sequence of defined operations, and produces an output. The algorithm must be clear, efficient, and unambiguous, ensuring that it can be implemented and executed to achieve the desired result.

Mathematical Representation of EdDSA

EdDSA is based on elliptic curve cryptography using **Edwards curves** and is typically implemented with the curve **Curve25519**. The algorithm operates in two main phases: **key generation** and **signature generation and verification**.

1. Key Generation

EdDSA key generation involves creating a private key and a corresponding public key. The steps are as follows:

- **Private Key (sk):** The private key is a randomly generated 256-bit value, represented as:

$$sk = r \quad (\text{where } r \text{ is a random integer modulo the curve order})$$

- **Public Key (pk):** The public key is derived from the private key using the elliptic curve scalar multiplication. The public key pk is computed as:

$$pk = r \cdot B$$

where B is a base point on the elliptic curve and r is the private key.

2. Signature Generation

To sign a message m , the private key sk and the message are used in the following steps:

- **Hash the Message:** The message m is hashed using a cryptographic hash function (e.g., SHA-512) to produce a 512-bit hash value:

$$H = \text{Hash}(m)$$

- **Compute the Signature:** The signature consists of two components, r and s . The calculation for these is as follows:

- First, compute H_1 , the hash of the message concatenated with the public key:

$$H_1 = \text{Hash}(pk \parallel H)$$

- Then, compute the signature components r and s :

$$r = (H_1 \cdot B) \quad s = (r + H_1 \cdot sk) \pmod{q}$$

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where q is the order of the base point B .

The resulting signature is the pair (r, s) , where r is the elliptic curve point and s is the scalar value derived from the private key and the hash of the message.

3. Signature Verification

To verify a signature (r, s) for a message m and a public key pk , the verifier performs the following steps:

- **Hash the Message:** Compute the same hash of the message as was done during signature generation:

$$H = \text{Hash}(m)$$

- **Compute the Hash of the Public Key and Message:**

$$H_1 = \text{Hash}(pk \parallel H)$$

- **Verify the Signature:** The verification checks if the following equation holds:

$$s \cdot B = r + H_1 \cdot pk$$

If the equation holds true, the signature is valid, indicating that the message has not been tampered with and was signed by the holder of the private key.

Summary of EdDSA Steps:

1. Key Generation:

- Private key sk is a random value.
- Public key $pk = r \cdot B$, where r is the private key and B is the base point.

2. Signature Generation:

- Hash the message m to get H .
- Compute $H_1 = \text{Hash}(pk \parallel H)$.
- Calculate $r = H_1 \cdot B$.
- Calculate $s = (r + H_1 \cdot sk) \bmod q$.

3. Signature Verification:

- Compute $H_1 = \text{Hash}(pk \parallel H)$.
- Verify $s \cdot B = r + H_1 \cdot pk$.

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8. SYSTEM STUDY AND TESTING

8.1 Feasibility Study

The feasibility of the project is analysed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ◆ Economical feasibility
- ◆ Technical feasibility
- ◆ Social feasibility

Economical Feasibility

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

Technical Feasibility

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

Social Feasibility

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The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

System Testing

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

8.2 Types of Tests

8.2.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

8.2.2 Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were

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individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

8.2.3 Functional testing

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

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Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

8.2.4 White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

8.2.5 Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box. you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

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- **TEST CASES:**

Input	Output	Result
Input text files	HEISENBERG GROUP-BASED DIGITAL SIGNATURES FOR SECURE COMMUNICATIONS	Success

- **Test cases Model building:**

S.NO	Test Cases	I/O	Expected O/T	Actual O/T	P/F
1	Register User	Enter username, email, password, and other details.	Registration successful.	Registration successful.	P
2	Login User	Enter username and password.	Login successful.	Login successful.	P
3	Login User	Enter invalid username or password.	Login failed.	Login failed.	P
4	Upload Document	Select document (file/video/music) and upload.	Document uploaded successfully.	Document uploaded successfully.	P
5	Upload Invalid File	Select invalid file type or no file selected.	Upload failed.	Upload failed.	P
6	Generate Heisenberg Signature	Select document to encrypt and sign.	Document signed successfully.	Document signed successfully.	P
7	Share Document	Enter recipient details and select document to share.	Document shared successfully.	Document shared successfully.	P

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S.NO	Test Cases	I/O	Expected O/T	Actual O/T	P/F
8	View Shared Document	Enter recipient credentials and select shared document.	Document displayed.	Document displayed.	P
9	Decrypt Document	Enter decryption key for document.	Document decrypted successfully.	Document decrypted successfully.	P
10	Verify Document Integrity	Upload document and check Heisenberg signature.	Signature verified successfully.	Signature verified successfully.	P
11	Logout User	Click on logout.	Logout successful.	Logout successful.	P

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9. CONCLUSION

In this project, a secure cloud storage system was developed using **EddSA (Edwards-Curve Digital Signature Algorithm)** to provide robust data protection. By leveraging the non-commutative properties of **EddSA**, the system ensures data integrity, authenticity, and confidentiality, making it resistant to both traditional and emerging cryptographic attacks, including those posed by quantum computing. The solution not only addresses current security challenges but also prepares for future threats, offering enhanced security and scalability. With faster signature generation and verification, the system ensures efficient performance while maintaining high security. Additionally, the user-friendly interface enables seamless data upload, retrieval, and management. Overall, this project provides a secure, scalable, and quantum-resistant solution for cloud storage, ensuring the long-term protection of sensitive data in an increasingly digital world.

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10. FUTURE ENHANCEMENT

Several enhancements can be introduced to improve the security, scalability, and usability of the cloud storage system. Adding **Multi-Factor Authentication (MFA)** would provide an extra layer of security during user login and document access. **Role-Based Access Control (RBAC)** could enable administrators to manage access to documents based on user roles, ensuring that sensitive data is restricted to authorized users. Implementing a **file versioning** system would allow users to track and revert to previous versions of uploaded files, especially useful in collaborative environments. Combining **EddSA** with symmetric encryption algorithms like **AES** for **hybrid encryption** could enhance performance by enabling faster file encryption while maintaining strong security. **Audit logging** would track all document activities for accountability and transparency. Additionally, integration with multiple cloud providers such as AWS, Google Cloud, and Azure would offer users flexibility in secure data storage. Lastly, **performance optimization** techniques would ensure that the system can handle large-scale data and a high number of concurrent users without compromising security or efficiency.

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